

Chain Rule and the Tree Diagram

Trim Ch 11, S. 12.6, 12.8

COMPOSITION

Recall the chain rule for one variable:

If $y = f(u)$ and $u = f(t)$

$$\Rightarrow y = f(u(t))$$

And the derivative of y with respect to t is written as,

$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dt}$$

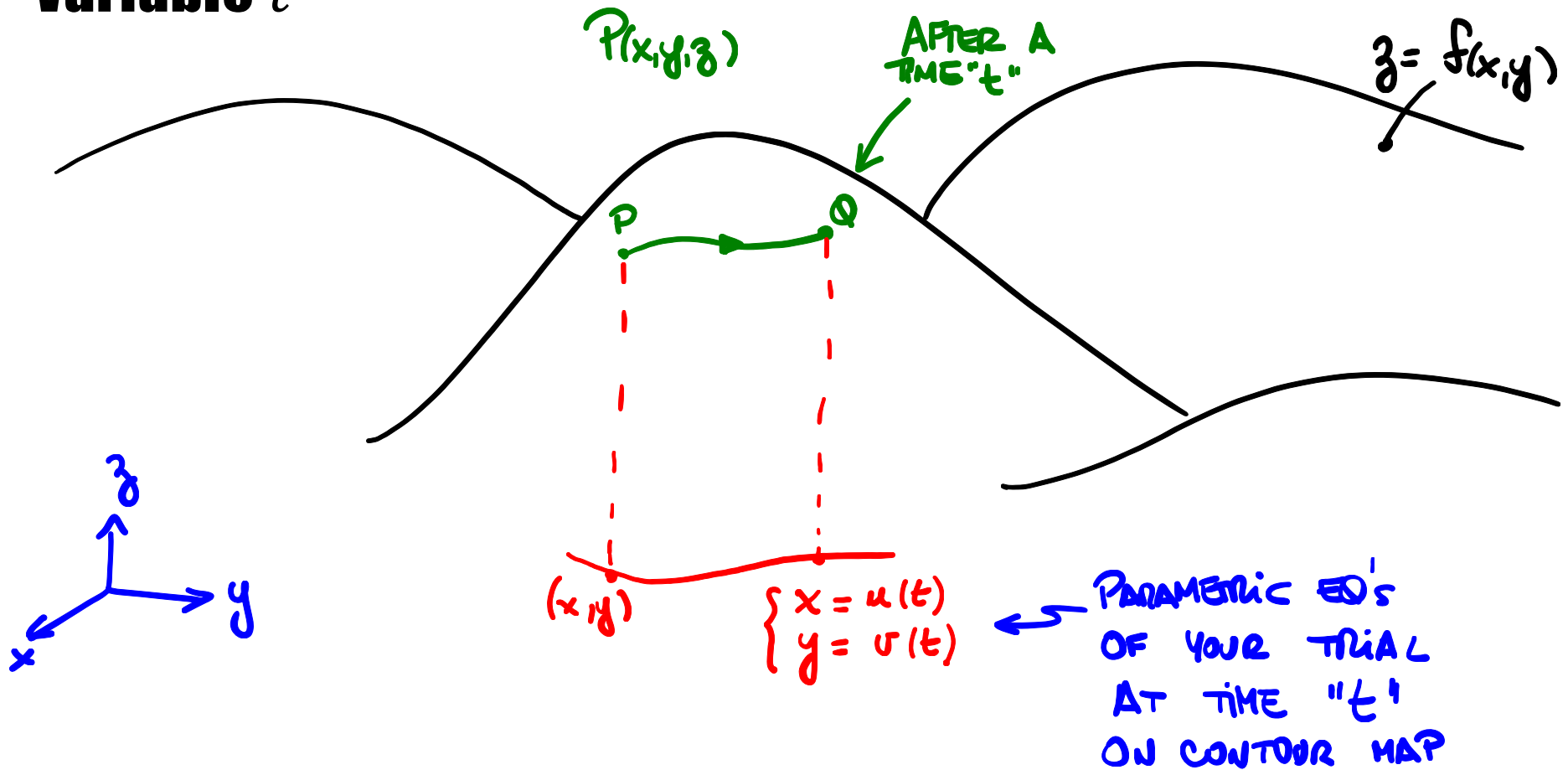
Outline:

1. Chain rule for multivariable functions
2. Chain Rule and the tree diagram



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Theorem 1: Chain Rule of $f(x, y)$ and one independent variable t



We apply the chain rule when our multivariable function is composed of inner functions that change over time and space.



Theorem 1: Chain Rule of $f(x, y)$ and one independent variable t

At time “ t ”, the altitude above sea level, $z = f(x, y)$, is given by the composite function,

$$z = f(\underbrace{u(t)}_x, \underbrace{v(t)}_y)$$

INTERMEDIATE VAR. INDEPENDENT VAR.

Question:

If you were hiking on these mountains, how fast would your altitude be changing with respect to time?

$$\frac{dz}{dt} = \frac{d}{dt} \left(f(u(t), v(t)) \right)$$

Theorem 1:

Using Leibnitz notation, the chain rule for differentiating this composition will be:

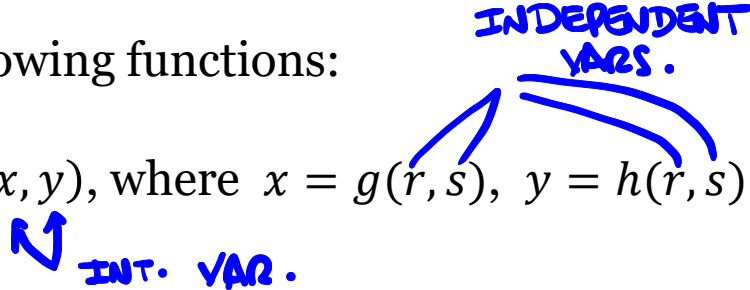
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$



Theorem 2: Chain Rule of $f(x, y)$ and two independent variables r and s

Let us consider the following functions:

$$\omega = f(x, y), \text{ where } x = g(r, s), y = h(r, s)$$



We can form the following composite function:

$$\omega = f(g(r, s), h(r, s))$$

Theorem 2:

The chain rule for differentiating this composition using Leibnitz notation will be:

$$\frac{\partial \omega}{\partial r} = \frac{\partial \omega}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial \omega}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial \omega}{\partial s} = \frac{\partial \omega}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial \omega}{\partial y} \cdot \frac{\partial y}{\partial s}$$

Notice that Theorem 2 has **one term for each intermediate variable** and each of these terms resemble the 1-D Chain rule.



Example:

Express $\frac{\partial \omega}{\partial r}$ and $\frac{\partial \omega}{\partial s}$ in terms of r and s if,

$$\omega = \underbrace{x^2 + y^2}_{f(x,y)}, \quad x = \underbrace{r - s}_{u(r,s)}, \quad y = \underbrace{r + s}_{v(r,s)}$$

Solution:

$$\begin{aligned} \frac{\partial \omega}{\partial r} &= \frac{\partial \omega}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial \omega}{\partial y} \cdot \frac{\partial y}{\partial r} \\ &= (2x)(1) + 2y(1) \\ &= 2x + 2y = 2(r-s) + 2(r+s) \\ &= 4r \end{aligned}$$

$$\frac{\partial \omega}{\partial s} = \frac{\partial \omega}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial \omega}{\partial y} \cdot \frac{\partial y}{\partial s}$$



Example:

$$= (2x)(-1) + (2y)(1)$$

$$= -2x + 2y$$

$$= -2(r-s) + 2(r+s)$$

$$= 4s$$



Chain Rule and the Tree Diagram

What if we have a composition involving differentiable functions of " n " intermediate variables?

$$f = f(x_1, x_2, \dots, x_i, \dots, x_n)$$

Where each one of the intermediate variable x_i are a function of " m " independent variables as,

$$x_i = g(t_1, t_2, \dots, t_j, \dots, t_m)$$

Then for any independent variable t_j we will have that,

$$\frac{\partial f}{\partial t_j} = \frac{\partial f}{\partial x_1} \cdot \frac{\partial x_1}{\partial t_j} + \frac{\partial f}{\partial x_2} \cdot \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial f}{\partial x_n} \cdot \frac{\partial x_n}{\partial t_j}$$

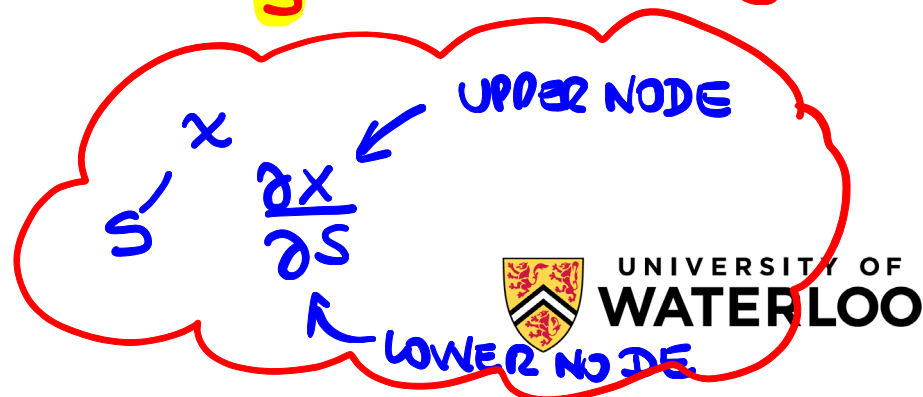
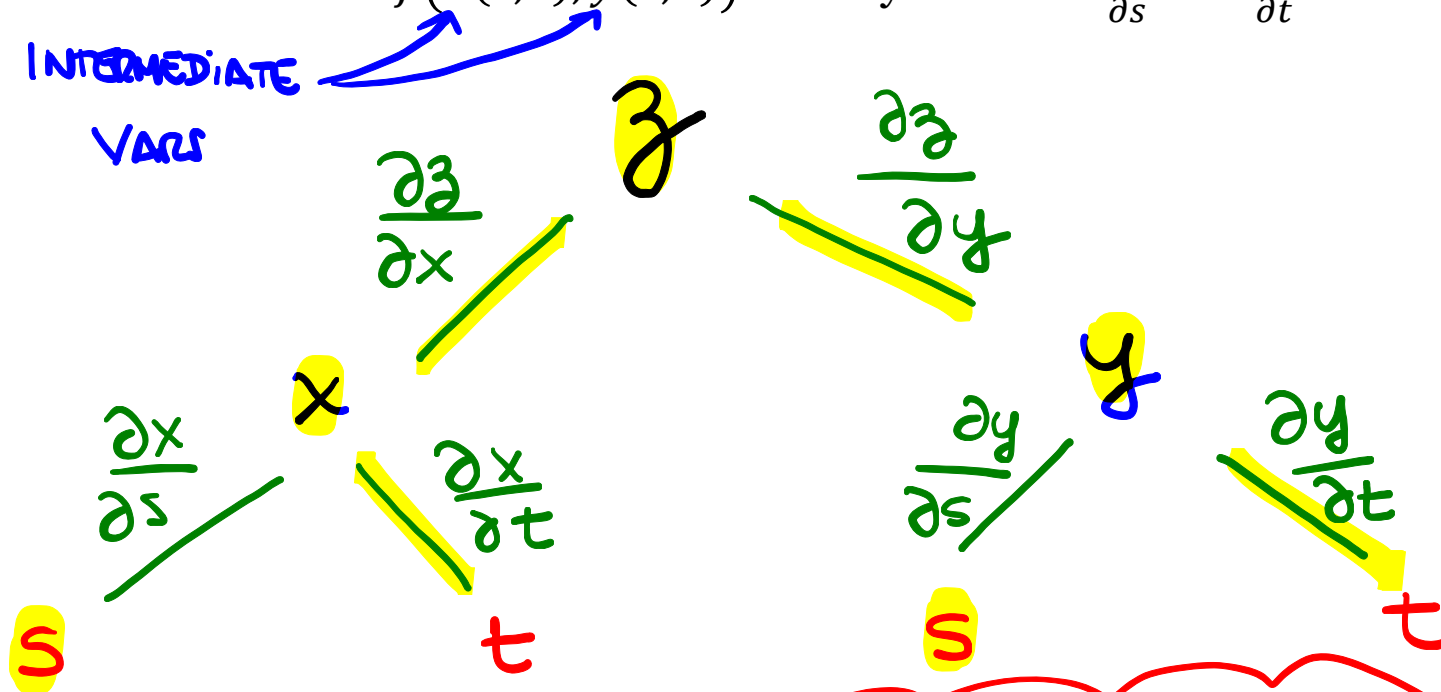


$$\frac{\partial f}{\partial t_m}$$



Chain Rule and the Tree Diagram

The tree diagram can be explained better using an example. To do this, let us look at the composite function $z = f(x(s, t), y(s, t))$ and try and find $\frac{\partial f}{\partial s}$ and $\frac{\partial f}{\partial t}$ *



* Notice that this is a **Theorem 2** case



Chain Rule and the Tree Diagram

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

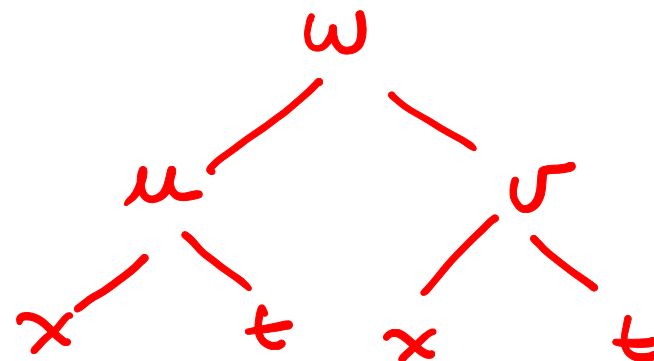
For our problem in previous lecture,

$$\omega = f(x - ct) + g(x + ct)$$

And our changes of variable,

$$u = x - ct$$

$$v = x + ct$$



The first partial derivative of ω with respect to t is,

To Construct the Tree Diagram ...

- We start at the top with the function itself (z)
- We branch out writing each intermediate variable
- We then out again and write the independent variables
- We connect the variables with one-to-one lines.
 - Each line represents a partial derivative

To Use the Tree Diagram ...

- **Start at the bottom of the tree diagram** and identify all branches that contain the independent variable you are differentiating with respect to.
 - **For each identified branch**, trace upward from the variable to the top of the tree, following the path along the branches.
 - **Multiply the partial derivatives** along each path from top to bottom.
 - **Repeat this process** for every branch that includes the same independent variable.
 - **Add together all the products** obtained from each path involving the independent variable.
- **Repeat the entire procedure for each independent variable** involved in the function.

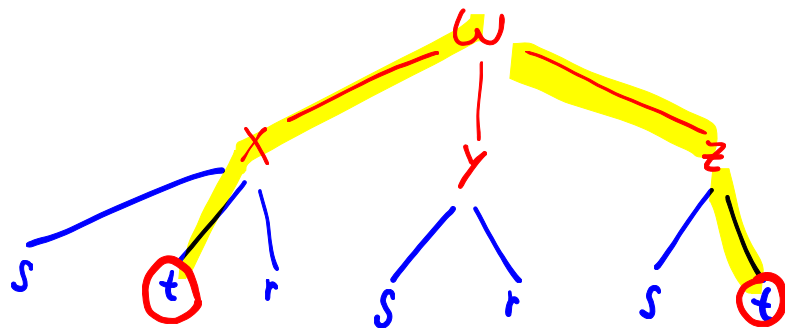
Example:

Use the tree diagram to write the chain rule for,

$$\frac{\partial \omega}{\partial t}$$

when $\omega = f(x, y, z)$, and $x = g_1(s, t, r)$, $y = g_2(s, r)$, $z = g_3(s, t)$

Solution:



$$\frac{\partial \omega}{\partial t} = \frac{\partial \omega}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial \omega}{\partial z} \cdot \frac{\partial z}{\partial t}$$



Summary

- The chain rule is a powerful tool to solve problems involving partial derivatives.
- The tree diagram help us solve partial derivatives of compositions involving multivariable functions.
- With some practice you should be able to use the chain rule to differentiate any composition of functions.



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Thank you